EXP 3: ID3 Decision Tree

CODE:

```
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder
import pandas as pd
data = pd.DataFrame({
   'Outlook': ['Sunny','Sunny','Overcast','Rain','Rain','Rain',
            'Overcast','Sunny'],
   'Humidity': ['High','High','High','High','Normal','Normal',
             'Normal','High'],
   'Play': ['No','No','Yes','Yes','Yes','No','Yes','No']
})

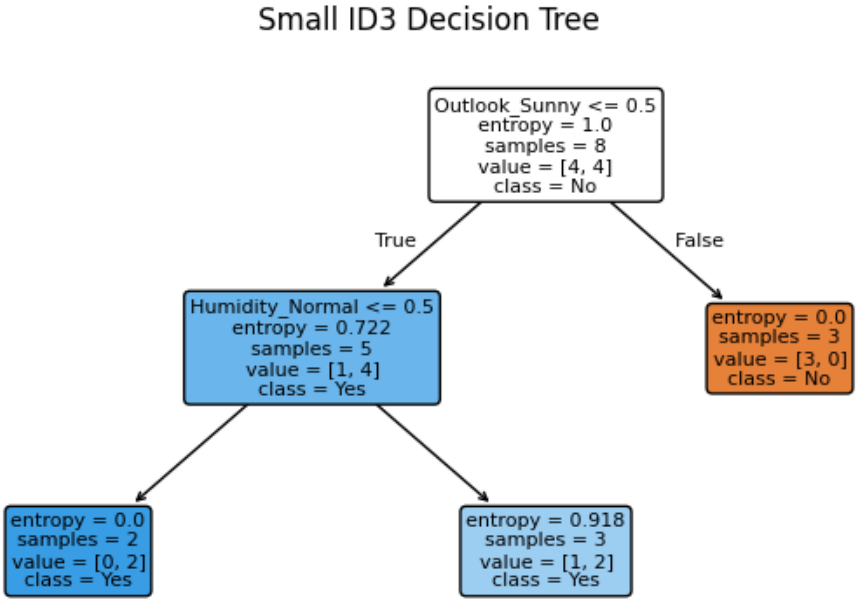
X = pd.get_dummies(data[['Outlook', 'Humidity']])
y = data['Play']

model = DecisionTreeClassifier(criterion='entropy', max_depth=2,
                        random_state=0)
model.fit(X, y)

plt.figure(figsize=(7, 4))
plot_tree(model, feature_names=X.columns,
        class_names=model.classes_,
        filled=True, rounded=True, fontsize=8)
plt.title("Small ID3 Decision Tree")
plt.tight_layout()
plt.show()
```

```
sample = pd.DataFrame([{'Outlook': 'Sunny', 'Humidity': 'High'}])
sample = pd.get_dummies(sample).reindex(columns=X.columns, fill_value=0)
print("Prediction:", model.predict(sample)[0])
```

OUTPUT



Prediction: No